

I

$$(1) \quad v_0 = \sqrt{2gh}, \quad t_0 = l \sqrt{\frac{2}{gh}}$$

$$(2) \quad |u| = \frac{|1-a|}{1+a} v_0, \quad |v| = \frac{2}{1+a} v_0$$

$$(3) \quad \frac{2(1+a)}{3-a} \frac{l}{v_0}$$

$$(4) \quad |u'| = \frac{|a^2 - 6a + 1|}{(1+a)^2} v_0, \quad |v'| = \frac{4|1-a|}{(1+a)^2} v_0$$

(5) A : 速さ v_0 のまま右に進む

B : 衝突のたびに速さが $2v_0$ ずつ増加し壁と A の間を往復する。

$$(6) \quad \frac{40}{9} \frac{l}{v_0}$$

II

$$(1) P_1 = P_i = P_0 + \rho d g, \quad P_3 = P_0$$

$$(3) \frac{P_0}{P_0 + \rho d g} \cdot \frac{h+d+l}{h} \quad \text{ist}$$

$$(5) Q_{12} = \frac{5}{2} (P_0 + \rho d g) S l$$

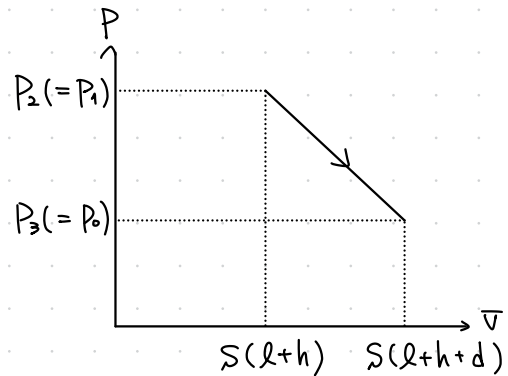
$$(6) P = P_0 + \rho \left(l + d + h - \frac{l}{S} \right) g$$

$$(7) W_{23} = (P_0 + \frac{1}{2} \rho g d) S d$$

$$(8) Q_{23} = \frac{5}{2} P_0 S d + \frac{1}{2} \rho d g S \{ d - 3(l+h) \}$$

$$(2) \frac{h+l}{h} \quad \text{ist}$$

$$(4) W_{12} = (P_0 + \rho d g) S l$$



III

(1) $\frac{2E}{R}$

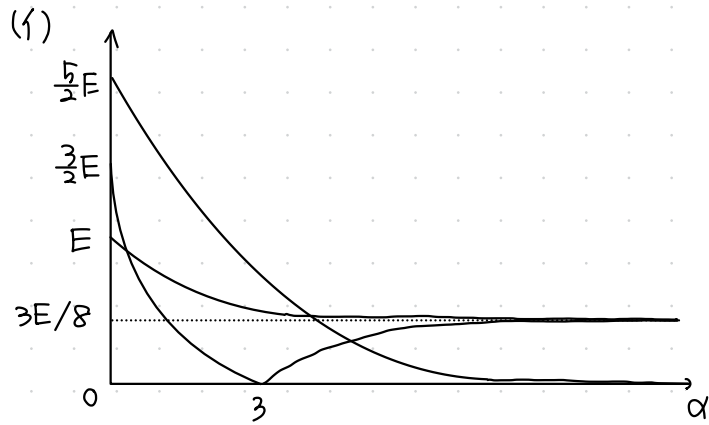
(2) $\frac{3}{2}E$

(3) $\frac{5}{2}E$

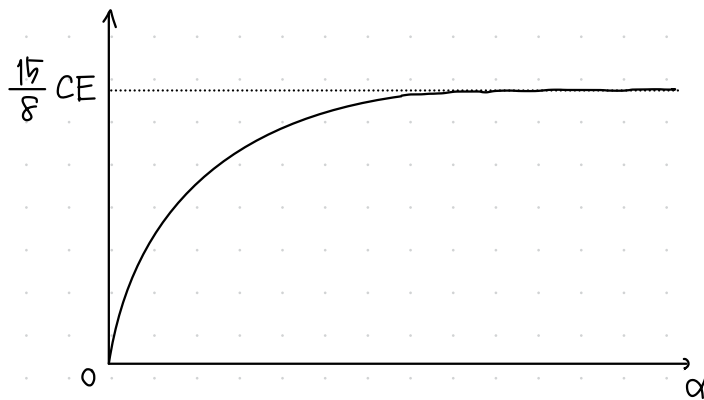
(4) (ア) $V_{c_1} = \frac{15}{2(3+4\alpha)} E$

$V_{c_2} = \frac{3(2+\alpha)}{2(3+4\alpha)} E$

$V_{c_1} = \frac{3|3-\alpha|}{2(3+4\alpha)} E$



(ウ) $\frac{15\alpha}{2(3+4\alpha)} CE$



(エ) $\frac{25}{54} CE^2$

